

Analytics Findings and Business Insights

Online Retail II Data Pipeline Project

Overview

After building the data pipeline and the analytics layer, I used the cleaned dataset to analyze revenue patterns and understand how the business behaves across time, countries, products, and customers.

The analytics model is based on a star schema. The central table is `analytics.fact_sales`, which stores the transactional data. It connects to several dimensions, including `dim_date`, `dim_country`, `dim_product`, and `dim_customer`.

Using this structure made the analysis much easier because I could aggregate and filter the data across multiple dimensions without repeatedly transforming the dataset.

The total net revenue (including returns) in the dataset is:
19,528,723.01

From there, I focused on identifying patterns in revenue, seasonality, geographic concentration, and customer behavior.

Seasonality in Revenue

One of the first things I noticed when plotting revenue over time was a clear seasonal pattern. Revenue consistently increases during the fourth quarter (October–December). The highest revenue levels appear in November and December, which aligns with the typical holiday shopping period in retail.

At the beginning of the year, especially in January and February, revenue drops noticeably. This likely reflects the post-holiday season slowdown.

I also noticed that December 2011 appears unusually low, but that is because the dataset ends early in that month rather than representing a real decline in sales.

Overall, this seasonal pattern confirmed that the date dimension and revenue calculations were working correctly in the analytics model.

Returns and Their Impact on Revenue

The dataset includes returns and cancellations, which appear as negative quantities.

Instead of removing these records, I decided to keep them because they represent real financial transactions. Removing them would have produced an unrealistic view of revenue.

The total impact of returns across the dataset is:
-1,103,155.26

Although this is a significant amount, it represents a normal level of returns for a retail dataset. Keeping these records allows the model to support both gross revenue analysis and net revenue reporting.

Geographic Revenue Distribution

When I analyzed revenue by country, I found that sales are heavily concentrated in a few markets.

The United Kingdom generates the largest share of total revenue. Other countries that contribute meaningful revenue include:

- EIRE
- Netherlands
- Germany
- France

This shows that the business relies heavily on a single primary market. From a business perspective, that kind of concentration can create risk if demand in that region changes.

Some smaller countries even show very low or slightly negative net revenue once returns are taken into account. This is not a modeling issue — it simply reflects real transactional behavior in the dataset.

Because the country is modeled as a separate dimension, comparing markets and aggregating revenue geographically is straightforward.

Product and Non-Merchandise Transactions

While analyzing product-level revenue, I noticed that not all high-revenue items represent physical products.

For example, items such as POSTAGE and DOTCOM POSTAGE appear frequently and contribute to revenue totals. These represent operational charges rather than merchandise sales.

There are also occasional records related to adjustments or operational corrections.

This means that, depending on the analysis's goal, these records may be included or excluded.

For financial reporting, they should remain in the data, but for product performance analysis, they might be filtered out.

The current analytics structure makes both approaches possible without changing the warehouse model.

Customer Revenue Distribution

Another pattern I observed is that revenue is not evenly distributed across customers.

A relatively small number of customers account for a large share of total revenue, while most contribute smaller amounts. This type of skewed distribution is common in retail data.

Because customer information is stored in a dedicated dimension, the model could easily support further analysis, such as:

- customer lifetime value
- repeat purchase behavior
- customer segmentation

These areas could be explored in future work.

Business Implications

Beyond the technical analysis, the results also suggest several practical insights.

First, the strong increase in Q4 demand means that businesses should expect higher transaction volumes during the holiday season. Inventory planning, logistics capacity, and staffing would likely need to increase during this period.

Second, the high revenue concentration in the United Kingdom suggests the business may be highly dependent on that market. Expanding activity in other regions could potentially reduce that dependence.

Finally, the seasonal increase in revenue does not appear to be caused by a few extreme transactions. Instead, the entire distribution of daily revenue shifts upward in Q4, suggesting that customer demand increases across the board during that period.

Conclusion

After building the pipeline, cleaning the data, and validating the transformations, the analytics layer produced results that are consistent with real retail behavior.

The star schema structure made it easy to analyze the data across multiple dimensions while keeping the warehouse model stable and reproducible.

Overall, this project demonstrates how an end-to-end data pipeline can support reliable analytics, business insight generation, and downstream applications such as dashboards and reporting.